

Pooya Khorrami

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Education

University of Illinois at Urbana Champaign

PhD Electrical and Computer Engineering

Advisor: Thomas S. Huang

Thesis Title: "How Deep Learning Can Help Emotion Recognition"

Urbana, IL

Jan. 2014–May 2017

University of Illinois at Urbana Champaign

MS Electrical and Computer Engineering

Advisor: Thomas S. Huang

Urbana, IL

Aug. 2011–Dec. 2013

Carnegie Mellon University

BS Electrical and Computer Engineering

Minor: Business Administration

Pittsburgh, PA

Aug. 2007–May 2011

Professional Experience

MIT Lincoln Laboratory

Technical Staff, AI Technology and Systems Group

Lexington, MA

Aug. 2017–Present

MIT Lincoln Laboratory

Summer Research Intern, AI Technology and Systems Group

Lexington, MA

June 2015–Aug. 2015

MIT Lincoln Laboratory

Summer Research Intern, AI Technology and Systems Group

Lexington, MA

June 2014–Aug. 2014

GE Aviation Systems

EID Intern

Grand Rapids, MI

June 2011–Aug. 2011

Carnegie Mellon University

Undergraduate Research Assistant

Pittsburgh, PA

June 2010–Aug. 2010

Technosciences Inc.

Student Intern

Beltsville, MD

June 2009–Aug. 2009

Technosciences Inc.

Student Intern

Beltsville, MD

June 2008–Aug. 2008

Publications

- [1] Bengt J Borgström and **Pooya Khorrami**. Adaptive score calibration for content-based image retrieval. In *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8612–8616. IEEE, 2026.
- [2] Josué Martínez-Martínez, Olivia Brown, Giselle Zeno, **Pooya Khorrami**, and Rajmonda Caceres. From snow to rain: Evaluating robustness, calibration, and complexity of model-based robust training. *arXiv preprint arXiv:2601.09153*, 2026.
- [3] **Pooya Khorrami**, Bengt J Borgström, Yunbin Deng, Charlie Dagli, and Pedro Torres-Carrasquillo. Constrained maximum likelihood gaussian score fusion for multimodal deepfake

- detection. In *2025 13th International Workshop on Biometrics and Forensics (IWBF)*, pages 1–6. IEEE, 2025.
- [4] Yunbin Deng, Alec Laprevotte, Robert Dunn, **Pooya Khorrami**, Charlie Dagli, and Pedro Torres-Carrasquillo. Emergent capability in audio deepfake detection. In *2025 13th International Workshop on Biometrics and Forensics (IWBF)*, pages 1–6. IEEE, 2025.
 - [5] Danielle Sullivan-Pao, Nicole Tian, and **Pooya Khorrami**. Lorax: Lora expandable networks for continual synthetic image attribution. In *BMVC Workshop on Media Authenticity in the Age of Artificial Intelligence (MAAAI)*, 2024.
 - [6] Andrew Alini, Douglas E Sturim, Kevin Brady, and **Pooya Khorrami**. Parasite networks: Transfer learning in resource-constrained domains. In *NeurIPS 2024 Workshop on Fine-Tuning in Modern Machine Learning: Principles and Scalability*, 2024.
 - [7] Ileana Rugina, Rumen Dangovski, Mark Veillette, **Pooya Khorrami**, Brian Cheung, Olga Simek, and Marin Soljacic. Meta-learning and self-supervised pretraining for storm event imagery translation. In *2023 IEEE High Performance Extreme Computing Conference (HPEC)*, pages 1–9. IEEE, 2023.
 - [8] Megan Frisella, **Pooya Khorrami**, Jason Matterer, Kendra Kratkiewicz, and Pedro Torres-Carrasquillo. Quantifying bias in a face verification system. In *Computer Sciences & Mathematics Forum*, volume 3, page 6. MDPI, 2022.
 - [9] **Pooya Khorrami**, Olga Simek, Brian Cheung, Mark Veillette, Rumen Dangovski, Ileana Rugina, Marin Soljacic, and Pulkit Agrawal. Adapting deep learning models to new meteorological contexts using transfer learning. In *2021 IEEE International Conference on Big Data Workshops (Big Data)*, pages 4169–4177. IEEE, 2021.
 - [10] Ileana Rugina, Rumen Dangovski, Mark Veillette, **Pooya Khorrami**, Brian Cheung, Olga Simek, and Marin Soljačić. Meta-learning and self-supervised pretraining for real world image translation. *arXiv preprint arXiv:2112.11929*, 2021.
 - [11] Kevin Brady, **Pooya Khorrami**, Lars Gjestebj, and Laura Brattain. Instance segmentation of neuronal nuclei leveraging domain adaptation. In *2021 IEEE High Performance Extreme Computing Conference (HPEC)*, 2021.
 - [12] **Pooya Khorrami**, Kevin Brady, Mark Hernandez, Lars Gjestebj, Sara Nicole Burke, Damon Lamb, Matthew A. Melton, Kevin Otto, and Laura Brattain. Deep learning-based nuclei segmentation of cleared brain tissue. In *2019 IEEE High Performance Extreme Computing Conference (HPEC)*, 2019.
 - [13] Richard Lippmann, Swaroop Vattam, **Pooya Khorrami**, and Cagri Dagli. A new data corpus to promote more complete autonomous machine learning pipelines. *2018 Neural Information Processing Systems Workshop (NeurIPSW)*, 2018.
 - [14] Prajit Ramachandran, Tom Le Paine, **Pooya Khorrami**, Mohammad Babaeizadeh, Shiyu Chang, Yang Zhang, Mark A Hasegawa-Johnson, Roy H Campbell, and Thomas S Huang. Fast generation for convolutional autoregressive models. *2017 International Conference on Learning Representations (ICLRW)*, 2017.
 - [15] Tom Le Paine, **Pooya Khorrami**, Shiyu Chang, Yang Zhang, Prajit Ramachandran, Mark A Hasegawa-Johnson, and Thomas S Huang. Fast wavenet generation algorithm. *arXiv preprint arXiv:1611.09482*, 2016.
 - [16] Kevin Brady, Youngjune Gwon, **Pooya Khorrami**, Elizabeth Godoy, William Campbell, Charlie Dagli, and Thomas S Huang. Multi-modal audio, video and physiological sensor

- learning for continuous emotion prediction. In *Proceedings of the 6th International Workshop on Audio/Visual Emotion Challenge*, pages 97–104. ACM, 2016.
- [17] James R Williamson, Elizabeth Godoy, Miriam Cha, Adrienne Schwarzentruher, **Pooya Khorrami**, Youngjune Gwon, Hsiang-Tsung Kung, Charlie Dagli, and Thomas F Quatieri. Detecting depression using vocal, facial and semantic communication cues. In *Proceedings of the 6th International Workshop on Audio/Visual Emotion Challenge*, pages 11–18. ACM, 2016.
- [18] Vuong Le, **Pooya Khorrami**, Usman Tariq, Hao Tang, and Thomas Huang. *Face Processing and Applications to Distance Learning*. World Scientific, 2016.
- [19] Wei Han, **Pooya Khorrami**, Tom Le Paine, Prajit Ramachandran, Mohammad Babaeizadeh, Honghui Shi, Jianan Li, Shuicheng Yan, and Thomas S Huang. Seq-nms for video object detection. *arXiv preprint arXiv:1602.08465*, 2016.
- [20] **Pooya Khorrami**, Tom Le Paine, Kevin Brady, Charlie Dagli, and Thomas S Huang. How deep neural networks can improve emotion recognition on video data. In *2016 IEEE International Conference on Image Processing (ICIP)*, pages 619–623, 2016.
- [21] **Pooya Khorrami**, Tom Le Paine, and Thomas Huang. Do deep neural networks learn facial action units when doing expression recognition? In *Proceedings of the IEEE International Conference on Computer Vision Workshops (ICCVW)*, pages 19–27, 2015.
- [22] Tom Le Paine, **Pooya Khorrami**, Wei Han, and Thomas S Huang. An analysis of unsupervised pre-training in light of recent advances. *2015 International Conference on Learning Representations (ICLRW)*, 2015.
- [23] Kai-Hsiang Lin, **Pooya Khorrami**, Jiangping Wang, Mark Hasegawa-Johnson, and Thomas S Huang. Foreground object detection in highly dynamic scenes using saliency. In *2014 IEEE International Conference on Image Processing (ICIP)*, pages 1125–1129, 2014.
- [24] **Pooya Khorrami**, Vuong Le, John C Hart, and Thomas S Huang. A system for monitoring the engagement of remote online students using eye gaze estimation. In *2014 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*, pages 1–6, 2014.
- [25] Shiyu Chang, Wei Han, Xianming Liu, Ning Xu, **Pooya Khorrami**, and Thomas S. Huang. Multimedia classification. *Data Classification: Algorithms and Applications*, page 337, 2014.
- [26] Thomas S. Huang, Vuong Le, Thomas Paine, **Pooya Khorrami**, and Usman Tariq. Visual media: History and perspectives. *IEEE MultiMedia*, 21(2):4–10, 2014.
- [27] **Pooya Khorrami**, Jiangping Wang, and Thomas S. Huang. Multiple animal species detection using robust principal component analysis and large displacement optical flow. In *ICPR Workshop on Visual Observation and Analysis of Animal and Insect Behavior (VAIB)*, 2012.

Technical Skills

Programming Languages (Proficient): Python, MATLAB

Programming Languages (Familiar): C/C++, C#

Libraries: PyTorch, Tensorflow, Scikit-Learn, Pandas, OpenCV, Theano, Lasagne, Caffe

Professional Activities

Associate Editor:

- IEEE Signal Processing Letters (SPL)

Program Committee Member:

- AAAI Conference on Artificial Intelligence (AAAI)
- International Joint Conference on Artificial Intelligence (IJCAI)
- IEEE Workshop on Analysis and Modeling of Faces and Gestures (AMFG)

Reviewer:

- AAAI Artificial Intelligence for Cybersecurity Workshop (AICS)
- ACM Multimedia RFIW Workshop (ACM MM RFIW)
- BMVC Workshop on Media Authenticity in the Age of Artificial Intelligence (MAAAI)
- Elsevier Pattern Recognition
- Elsevier Pattern Recognition Letters
- IEEE Access
- IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)
- IEEE Conference of the Engineering in Medicine and Biology Society (EMBC)
- IEEE Journal of Translational Engineering in Health and Medicine (JTEHM)
- IEEE Symposium on Technologies for Homeland Security (HST)
- IEEE Transactions on Emerging Topics in Computational Intelligence (TETCI)
- IEEE Transactions on Systems, Man, and Cybernetics: Systems (TSMCA)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Affective Computing (TAC)
- IEEE Transactions on Image Processing (TIP)
- IEEE Transactions on Information Forensics and Security (TIFS)
- IEEE Transactions on Multimedia (TMM)
- IEEE Signal Processing Letters (SPL)
- IEEE Winter Conference on Applications of Computer Vision (WACV)
- IEEE International Conference on Multimedia and Expo - FacesMM Workshop (ICMEW FacesMM)
- Workshop on Bringing Semantic Knowledge into Vision and Text Understanding (IJCAI-TUSION)

Conference / Workshop Organization:

- Co-organizer of Synthetic and Adversarial ForEnsics (SAFE) Workshop at WACV 2026
- Co-organizer of Synthetic and Adversarial ForEnsics (SAFE) Workshop at CVPR 2026
- Co-chair of Poster Session at ICASSP 2026

Awards and Honors

- R&D 100 Award (Co-recipient) - Lincoln Laboratory Radio Frequency Situational Awareness Model - 2025
- James Henderson Fellowship - 2012
- Andrew Carnegie Society Scholar - 2011
- Xerox Technical Minority Scholarship - 2009, 2010
- Carnegie Institute of Technology Dean's List - 2007-2012

Other Information

- U.S. Citizen